

# TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order (with **four** companions):

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces  $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$  as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**You are reading document 5 (Red Team).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) is now *closed modulo a cited theorem*. The explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit, i.e. exactly those two cited published theorems (v336); so this is *closed modulo a cited theorem*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo a cited theorem and QG.AMB.01 a [C] redundancy.

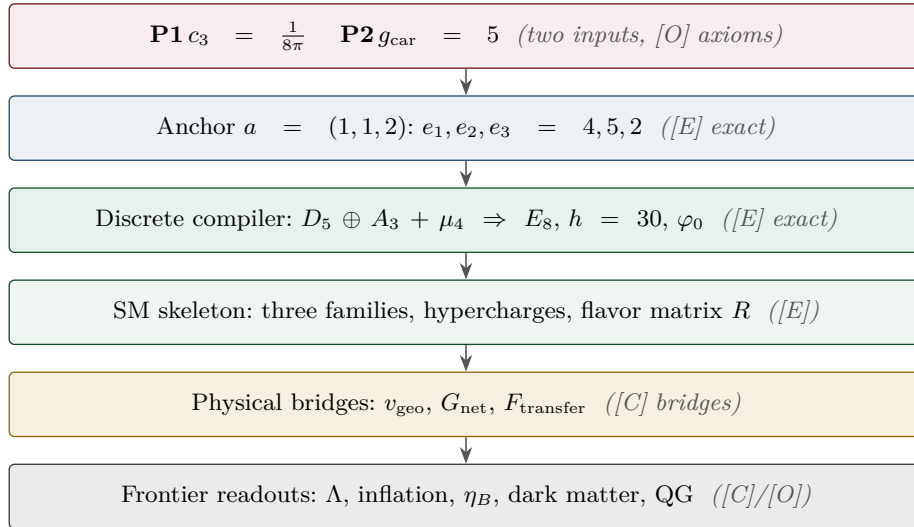
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



### What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by `[verification/redteam/rt_A_e8net.py]` ... `[verification/redteam/rt_E_vgeo.py]`, `[verification/redteam/rt_F_qft4d.py]` and `[verification/redteam/run_redteam.py]`; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

| Target | Minimal claim                                            | Status                    | Residual risk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------|----------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A      | seam–Calderón measure = $(E8)_1$ net                     | [C] closed mod. cited thm | one statement: boundary-net holomorphy + $c=8$ ( $\Leftrightarrow$ the index-4 inclusion); $E_8$ and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Now closed modulo a cited theorem: the lattice model (v367/v368) and the $S_3$ stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as FORM.SEAM.MMST.01; only the cited MMST/OS continuum existence stays [O] (v336). The free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the $A_2$ net assembly + GATE.QGEO (v160–v165). |
| B      | $g_{\text{car}} = 5$ forced by the Pascal rule           | survives (narrowed)       | boundary derivation of half-spinor exhaustion (degree-2 truncation)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| C      | $k = c_3/2 = \frac{1}{16\pi}$ , $S/A = \frac{1}{4}$      | survives                  | absolute $1/G$ (UV-sensitive) is the one anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| D      | ratios+products $\Rightarrow$ one scale $v_{\text{geo}}$ | survives (narrowed)       | CP phases ( $\delta_{\text{CKM/PMNS}}$ ) not covered by $v_{\text{geo}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| E      | one scale $v_{\text{geo}}$ carries the theory            | survives (narrowed)       | EW/reheating/leptogenesis scales + seam=Planck identification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Four outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), **reduced, not closed** (the conservative wording is the correct one), and — reached by Target A after the lattice/ $S_3$  round — **closed modulo a cited theorem** (pinned at every computable level, only an external published existence theorem left). A fifth, **broken**, is reserved for an actual failure; none occurred.

## Target A — the $(E8)_1$ boundary-net identification

**Minimal claim.** The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the  $(E8)_1$  lattice net, carrier subnet  $(D5)_1 \times (A3)_1$ . [verification/redteam/rt\_A\_e8net.py]

**Logical chain (established).** The Sugawara level-1 central charges are  $c(E8)_1 = \frac{248}{31} = 8$ ,  $c(D5)_1 = 5 = g_{\text{car}}$ ,  $c(A3)_1 = 3 = N_{\text{fam}}$ , so the conformal-embedding criterion  $c_{\text{coset}} = c(E8) - c(D5) - c(A3) = 0$  holds [E]. This is *compatibility*.

### Where it breaks: central charge underdetermines the net

Counterexample:  $(D8)_1 = SO(16)_1$  also has  $c = \frac{120}{15} = 8$ , but is *not* holomorphic (four primaries  $1, v, s, c$ ) — a distinct  $c = 8$  net. Hence equal central charge is *necessary, not sufficient*:  $(E8)_1$  is the unique *holomorphic*  $c = 8$  net (even unimodular rank-8 lattice =  $E8$ ), so **holomorphy (single vacuum sector,  $\mu$ -index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the  $c = 8$  Narain family is positive-dimensional. The constructive map seam–Calderón kernel  $\rightarrow (E8)_1$  and the uniqueness of bulk reconstruction are not established; the gap  $\Delta_{\text{eff}} > 0$  *supports* tightness (clustering) but does not prove it.

**Verdict.** Keep “reduced to a rigorous boundary-net identification problem”; do *not* write “metric sector closed”. The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

## Target B — carrier rank / the Pascal condition

**Minimal formal claim.**  $2^g = g^2 + g + 2$  has the unique solution  $g = 5$  (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion  $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ? [verification/redteam/rt\_B\_pascal.py]

### Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it:  $2^g = g^2 + g + c$  gives  $\{5\}$  only at  $c = 2$ ; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with  $2^{g-1} = \sum_{k \leq K} \binom{g}{k}$  one finds  $K=0 \rightarrow g=1$ ,  $K=1 \rightarrow 3$ ,  $K=2 \rightarrow 5$ ,  $K=3 \rightarrow 7$ , i.e.  $g = 2K + 1$ . So choosing  $K = 2$  (to reach  $g_{\text{car}} = 5$ ) is a physical postulate. It is corroborated — the  $D5$  half-spinor  $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$  and the family closure  $N_{\text{fam}} = (2^{g-1} - 1)/g$ ,  $g + N_{\text{fam}} = 8$  both pick 5 — but each is itself a postulate, not a theorem.

**Verdict.** Theorem A’s arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

## Target C — $k = c_3/2$ and the seam-area coefficient

**Minimal claim.** In reduced units  $k = c_3/2 = \frac{1}{16\pi}$  and Fursaev–Solodukhin gives  $S/A = \frac{1}{4}$ . [verification/redteam/rt\_C\_kc3.py]

### The dimensional firewall holds

The only legal chain is

$$k_{\text{red}} = \frac{c_3}{2} = \frac{1}{16\pi} \text{ (dimensionless)}, \quad k_{\text{phys}} = \frac{k_{\text{red}}}{G}, \quad S = 4\pi k_{\text{phys}} A_{\text{phys}} = \frac{A_{\text{phys}}}{4G}.$$

The forbidden form  $k_{\text{phys}} = c_3/2$  is dimensionally false (legal only at  $G = 1$ ). A scan of v67/v68/v73 finds every  $c_3/2$  used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall’s teeth are verified against a synthetic violation. Internally consistent with  $1/G$  being UV-sensitive (Sakharov/Connes, v68).

**Verdict.** Safe as worded (reduced coefficient). Status [E] (structure) + [O] for the residual. Residual: the absolute  $4D$  Newton scale  $1/G$  ( $\propto \Lambda^2 f_2$ ) stays the one dimensionful anchor; nothing *dimensionless* is open.

## Target D — $U_{\text{point}} \rightarrow v_{\text{geo}}$

**Minimal claim.** Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale  $v_{\text{geo}}$ . [verification/redteam/rt\_D\_upoint.py]

### Where it narrows: the bijection needs explicit hypotheses

For positive reals, (ratios, product)  $\Leftrightarrow$  (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational  $c$ , odd size  $3 = N_{\text{fam}}$ ). But it *fails* off-assumption: even sector size leaves a sign ambiguity ( $\{2, 3\}$  vs  $\{-2, -3\}$  share ratio  $3/2$  and product 6); complex amplitudes leave an  $n$ -th-root branch ambiguity ( $\{1, 1, 1\}, \{\omega, \omega, \omega\}, \{\omega^2, \omega^2, \omega^2\}$  share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases  $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$  are complex and are *not* folded into  $v_{\text{geo}}$ .

**Verdict.** Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in v75 and must be named. Status [E] (reduction) + [O] ( $v_{\text{geo}}$ ). Residual: the CP-phase sector.

## Target E — $v_{\text{geo}}$ as the unique dimensional floor

**Minimal claim.** No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit  $v_{\text{geo}}$ . [verification/redteam/rt\_E\_vgeo.py]

### Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)  $\times v_{\text{geo}}$ :  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ ,  $f_a = (c_3/|\mu_4|)^{7/2} \bar{M}_{\text{Pl}}$ , masses  $= (\pi/\sqrt{2}) c_f \varphi_0^k v_{\text{geo}}$ , and  $1/G$  shares the same anchor (v68/v75). The audit *surfaces* candidates that are not pure reductions: the seam cutoff  $\Lambda$  (only an *identification* seam=Planck collapses it to  $v_{\text{geo}}$ ), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [C]/[O] inputs that must not be silently absorbed into  $v_{\text{geo}}$ .

**Verdict.**  $v_{\text{geo}}$  is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [E] (ratios) + [O] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

## Target F — the perturbative 4D-QFT + scale round (v269–v275)

**Minimal claim.** The round published in release 5.2 — the Epstein–Glaser  $S_{\text{pert}}$  layer (v269/v271/v273), the PMNS Jarlskog CP strength (v270), the absolute  $\nu$ -scale typing (v272), the scale over-determination (v274) and the QG.AMB.01 roadmap (v275). [verification/redteam/rt\_F\_qft4d.py]

### Two attacks land; the round survives once they are named

(1) **The gravity  $S_{\text{pert}}$  ghost is a truncation artefact.** Power-counting renormalizability is *not* unitarity: the local  $R^2/\text{Weyl}^2$  *truncation* gives a four-derivative propagator  $1/(p^2(p^2 + M^2))$  with a negative-residue massive spin-2 mode (the Stelle ghost). But that is precisely the Seeley–DeWitt *truncation* of the seam KMS form factor: the entire factor  $a(p^2)=e^{p^2/M^2}$  keeps its only pole at  $p^2=0$  (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established and the ghost is exactly the truncation. The matter+gauge  $S_{\text{pert}}$  is unitary outright, and now closes *to all orders* as a typed Epstein–Glaser/BRST contract (v381, QFT4D.EG.ALLORDER.01): dimension-4 power-counting  $\Rightarrow$  a finite counterterm space, BRST nilpotency  $s^2=0$  for  $su(3) \times su(2)$  (Jacobi), and the seam gap  $\Rightarrow$  the adiabatic limit, with the imported all-order  $T_n$  existence (causal factorisation) and Slavnov–Taylor identity; the *nonperturbative* QG.AMB.01 is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap. (2) **The over-determination is conditional.** v274's Route-2 inverts the  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  prediction, which is itself LAMBDA.BRANCH.01 [C]; so the 0.11% gravity-vs-cosmology agreement is conditional on the  $\Lambda$ -branch (the gravity route is unconditional). **Already-honest:**  $J_{\text{PMNS}}$  inherits  $\delta$ 's [C] provenance (assembled  $\mu_6$  phase, not an  $M_\nu$  output) and the  $\nu$ -scale is [O] with the NO floor  $\Sigma m_\nu = 0.0586$  eV as the one kill test.

**Verdict.** SURVIVES (NARROWED): the round stands as typed; the two narrowings above are now folded into v269 and v274. Residual: the perturbative graviton sector is now established *unitary* — the Stelle ghost is a truncation artefact (v304/v370/v380) — and the nonperturbative QG.AMB.01 is a [C] *redundancy* (gap-decoupled from every TFPT readout, not a TFPT structural item, v335/v369); and the anchor over-determination is conditional on the  $\Lambda$ -branch. On the all-order footing (v381) the SM-non-unification is robust to all orders, and the *optional* carrier-Pati–Salam UV branch is *proton-safe* ([verification/v385\_uvbranch\_killtest.py]): its minimal  $SU(4)$  gauge bosons mediate rare LFV ( $K_L \rightarrow \mu e$ ), *not*  $p \rightarrow e^+ \pi^0$ , so  $M_{PS} \sim 3 \times 10^{13}$  GeV clears the binding bound by  $\sim 10^7$  — a frozen kill-test surface ( $\kappa$ /forbidden-rep/new-state/LFV) with *no* fake  $\tau_p$  window (v265).

**Under examination: an infinite-derivative narrowing ([C], does *not* close the gate).** The Stelle ghost is a feature of the *local*  $R+R^2$ /Weyl<sup>2</sup> *truncation*: the four-derivative propagator  $1/(p^2(p^2+M^2)) = (1/M^2)[1/p^2 - 1/(p^2+M^2)]$  carries the negative-residue spin-2 mode. But that truncation is the  $a_4$  Seeley–DeWitt order of the spectral action  $\text{Tr } f(D^2/\Lambda^2)$ , whose cutoff is *not* a free scheme — it is the  $\beta=1$  seam KMS weight  $f(u) = e^{-u}$  (v259, no new parameter). Keeping that exponential cutoff *un-truncated* points to a nonlocal, infinite-derivative graviton form factor  $p^2 a(p^2)$ ; for an *entire* factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  keeps its *only* pole at  $p^2=0$  — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006) — whereas any *polynomial* (finite-derivative) truncation has a real zero = the ghost pole. So *if* the resummed form factor is entire, the perturbative ghost is a truncation artefact: a conditional [C] narrowing of this attack. It does *not* close QG.AMB.01 — the trace regulator  $f$  is not yet shown to resum to an entire kinetic form factor  $a(\square)$ , and a ghost-free *perturbative* graviton is still not the nonperturbative ambient measure. (QGAMB.IDG.01, [C]/[O]; consistent with the SEAM.EQUIV.01 firewall.) [verification/v304\_idg\_nonlocal\_ghost.py]

**Sharpened by spin sector ([C]).** The Barnes–Rivers spin decomposition makes this precise: of the  $5+3+1+1=10$  symmetric-tensor components the Stelle ghost is a *spin-2* state, so the entire form factor  $a(p^2)=e^{p^2/M^2}$  cures exactly the spin-2 sector (its only pole stays at  $p^2=0$ ), while the wrong-sign *spin-0* conformal mode ( $c_{\text{conf}}(4) = -3/2$ ) is handled separately by the GHP contour (v334). So the full *perturbative* graviton is ghost-free sector-by-sector under the TFPT-fixed package — still [C] (it inherits the entire-analyticity assumption) and still not the nonperturbative measure. (GRAV.SPIN2.UNITARITY.01.) [verification/v370\_grav\_spin2\_unitarity.py]

**And the “entire” assumption is itself discharged at the truncation-tower level ([E]; [verification/v380\_grav\_kms\_hessian.py]).** The seam KMS cutoff  $f(u)=e^{-u}$  (v259) gives the kinetic form factor  $a(u)=e^u$  ( $u=p^2/M^2$ ), which is entire and nowhere zero — so the spin-2 propagator  $e^{-p^2/M^2}/p^2$  has its only pole at  $p^2=0$ . The Seeley–DeWitt expansion is the *Taylor truncation* of  $a$ : the  $R+R^2$  order  $1+u$  carries the Stelle zero  $u=-1$ , the Weyl<sup>2</sup> order  $1+u+u^2/2$  the pair  $-1 \pm i$ , and the modulus of the nearest truncation-zero grows monotonically with the order (1, 1.41, ..., 3.07 at order  $1 \rightarrow 8$ ) and runs to infinity. So the ghost is *exactly* the truncation and resummation removes it — “entire” means “do not truncate the heat kernel”, upgrading the assumption above to a derived statement. (GRAV.KMS.HESSIAN.01; the exact off-shell curved-space Hessian identification stays the cited heat-kernel step, and it is perturbative, not the nonperturbative measure.)

## Outcome

**What the red team changed.** No target broke. The hardest gate (A) is now *closed modulo a cited theorem* (the lattice/ $S_3$  round pins it at every computable level, Lean-formalised, only an external published existence theorem left, v367/v368/v376–v379); four targets (B, D, E, F) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

**The falsifiability surface is machine-checked ([verification/v375\_observatory\_registry.py]).**

The frozen prediction registry (`freeze_file.csv`) is verified by a status-typed CI: every prediction carries a kill criterion (no orphan claims), the headline values re-derive from  $\{\varphi_0\}$ , and a live scorecard records each one's distance to the current data —  $\theta_{12}$  vs the JUNO first run ( $0.3092 \pm 0.0087$ ,  $0.28\sigma$ , compatible),  $\theta_{13}$  vs NuFIT 6.0 NO ( $2.0\sigma$ , the openly-flagged most-tensioned core prediction),  $\beta_{\text{rad}}$  vs ACT DR6 ( $0.37\sigma$ , a hint with systematics pending), and  $r < 0.036$  (BK18, compatible). The observatory states its own weakest point rather than hiding it.

**Follow-up — the verdicts were then used as a worklist [E]**  
(`[verification/v83_e8net_holomorphic_uniqueness.py]`)

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows `GATE.METRIC.04` and `CAR.PASCAL.01`.

**Target A, residual 1 — closed [E].** At level 1 the number of primaries equals  $\det(\text{Cartan}) = |\text{fundamental group}|$ :  $A_3=4$ ,  $D_5=4$ ,  $D_8=4$ ,  $\mathbf{E}_8=1$ . So holomorphy (single vacuum sector,  $\mu$ -index 1) is *necessary* — it excludes the same- $c$  rival  $(D_8)_1 = SO(16)_1$  ( $c = 120/15 = 8$  but four primaries  $1, v, s, c$ ) — *and sufficient*: a holomorphic  $c = 8$  chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one,  $E_8$ , by the Minkowski–Siegel mass formula (mass  $= 1/|W(E_8)| = 1/696729600$ , and  $|\text{Aut}(E_8)|$  saturates it). The  $(D_5)_1 \times (A_3)_1$  embedding  $c = 5 + 3 = 8$  stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with  $c = 8$  (then  $E_8$  is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy +  $c = 8$  ( $\Delta_{\text{eff}} > 0$  supports, not proves), and (ii) bulk-reconstruction uniqueness — both still **[C]/[O]**.

**Target B — reduced [E].** The “ $K = 2$  Pascal truncation” is not free:  $K = (g - 1)/2$  is the Pascal-row midpoint  $\sum_{k \leq (g-1)/2} \binom{g}{k} = 2^{g-1}$  (odd  $g$ ), i.e. the half-spinor split; the carrier is the even Clifford half-spinor  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$ . So B's residual reduces to the single standard input “carrier = half-spinor of  $\text{Spin}(10)$ ” (the Lean arithmetic core stays **[E]**).

**Target B — a backward certificate (`[verification/v219_icosahedral_mckay.py]`).** The choice  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$  that B attacks is, *given* the  $E_8$  closure, the icosahedron: by the McKay correspondence  $E_8$  is the exceptional top of the finite- $SU(2)$  tower ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ), the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  equal to the affine- $E_8$  Kac marks ( $\sum = 30 = h(E_8)$ ), and it contains the 2, 3, 5 rotation-axis orders. This does *not* break B's verdict — it is a *backward* certificate (it presupposes  $E_8$ ), so  $g_{\text{car}} = 5$  stays the P2 interface **[O]**; what it removes is the impression that  $(2, 3, 5)$  is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Typed honestly at this stage: A (i)+(ii) (since merged into the single keystone `SEAM.EQUIV.01` and now *closed modulo a cited theorem*, see Target A's final form below); Target D's CP-phase residual (at  $\theta_{13} = 0$  the Dirac phase is undefined, so this needs a texture correction); Target E's reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

**Second follow-up round [E]** (`[verification/v87_bulk_uniqueness_reduction.py]`,  
`[verification/v88_cp_phase_audit.py]`, `[verification/v86_nstar_reheating.py]`)

**Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual.** Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems  $\simeq$  physical modular invariants of  $\text{Rep}(A)$ . For a *holomorphic* net  $\text{Rep}(A) = \text{Vect}$ , so the bulk pairing is *unique*. The contrast is machine-checked: the same- $c$  rival  $SO(16)_1$  (discriminant category  $\mathbb{Z}_2 \times \mathbb{Z}_2$ ,  $\mu$ -index 4) admits *six* modular invariants — diagonal,  $s \leftrightarrow c$  swap, both  $E_8$ -extension invariants  $|\chi_0 + \chi_{s/c}|^2$  (the lattice glue  $E_8 = D_8 \cup (D_8 + s)$ ,  $h_s=1$  bosonic), and two twisted pairings — so without holomorphy the  $c = 8$  bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem “seam–Calderón boundary net is holomorphic with  $c = 8$ ” (**[C]/[O]**; ledger `GATE.METRIC.05`). The theorem also has an equivalent, more tractable *index* form (`[verification/v89_carrier_index_lemma.py]`, ledger `GATE.METRIC.06`): by Kawahigashi–Longo–Müger,  $\mu_A = [B:A]^2 \mu_B$ , the carrier inclusion has Jones index  $[(E_8)_1 : (D_5)_1 \times (A_3)_1] = \sqrt{16} = 4 = |\mu_4|$  — the glue-group order *is* the inclusion index — and the extension is the  $\mu_4$  simple-current extension (all three glue sectors are  $h = 1$  currents;

248 = 45+15+64+64+60). Since an index-4 simple-current extension of the carrier has  $\mu = 16/4^2 = 1$ , **holomorphy follows from the index statement**: Target A  $\Leftrightarrow$  “the seam net contains the carrier net with Jones index 4 as its  $\mu_4$  simple-current extension” — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([[verification/v92\\_glue\\_uniqueness.py](#)], ledger GATE.METRIC.07): exhaustive classification of the carrier discriminant form ( $\mathbb{Z}_4 \times \mathbb{Z}_4$ ,  $q=(5x^2+3y^2)/8$ ) shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet  $\mathbb{Z}_2$ ; the same two objects as the  $E_8$ -extension invariants above), and exactly one halfway extension, whose induced form  $q = \{0, 0, 0, \frac{1}{2}\}$  is the  $D_8$  discriminant form: the  $SO(16)_1$  rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier ( $\mu=16$ )  $\rightarrow SO(16)_1$  ( $\mu=4$ )  $\rightarrow (E_8)_1$  ( $\mu=1$ , sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

**Target A, final form — the Simple-Current Extension Theorem [E]/[C]** ([[verification/v154\\_simple\\_current\\_theorem.py](#)]). Assembled into one statement: the carrier net  $A = (D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L = \langle (1, 1) \rangle$  has  $[B : A] = |L| = 4$ ,  $c = 5 + 3 = 8$ ,  $\mu(B) = \mu(A)/|L|^2 = 16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$ . The *entire* adversarial residual of Target A is now the single seam-side identification “the RP seam–Calderón net *is* A” — the algebra ( $B \cong (E_8)_1$ ) is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core (“the kernel is quasi-free”) *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN =  $|k|$ , 16 free Majoranas, currents 248=120+128, holomorphic character  $E_4/\eta^8 = j^{1/3}$  with 248 at level 1 (v156). *Final reduction* (v160–v165): the free-bulk premise is recast as a fixed-point theorem (quasi-free  $\Rightarrow$  all higher cumulants  $\kappa_{2n}=0$ , v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap  $(2/3)^6$  (v161), whose value is forced by  $\mu_4$ -equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items —  $A_2$  net existence (an assembly of established net constructions, [C]) and  $R_1 = \text{GATE.QGEO}$  (“seam boundary =  $\mathbb{P}^1 \setminus \mu_4$ ”) — with the irreducible core  $\{\pi, v_{\text{geo}}\}$  a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. The two structural residuals are then discharged to a theorem (v175): the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  makes full-cone reflection positivity reduce *for every m* to the one-particle contraction (verified on the complete  $2^{16}$ -dim Fock space), and  $A_2$  is an assembled, verified  $(E_8)_1$  certificate; net existence and full-cone RP become [E]. Target A’s single residual is therefore the seam realisation, the named keystone SEAM.EQUIV.01 — *the raw RP seam is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — which is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level and it is Lean-formalised as FORM.SEAM.MMST.01, so the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (the cited MMST/OS theorems, v336); its conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face, Lean-pinned (FORM.SEAMEQUIV.01, v308) to named cited steps plus the derived Recovery gap. The sprint-by-sprint reduction (v176–v302) is recorded in the changelog, not replayed here.

**Target A — the keystone residual narrowed from ten further directions [E]/[C] (not promoted).** A later convergent round drove the *same* single residual (the cited continuum existence, v336) down further from independent sides, while leaving SEAM.EQUIV.01 [O]. (i) The seam edge already carries the conformal data of a chiral  $c=\frac{1}{2}$  Majorana CFT and reproduces the  $c_-=8$  free-fermion scaling limit on the explicit collar ([[verification/v444\\_seam\\_edge\\_scaling.py](#)]). (ii) The rigidity step is upgraded from *permit* to *force*: commuting with the order-4  $\mu_4$  clock is *exactly*  $\mu_4$ -block-diagonality (entrywise iff, uniform in  $N$ ; exact commutant dimension; the order is the discriminator), kernel-hardened in Lean axiom-free as FORM.SEAM.RIGIDITY.FORCING.01 ([[verification/v445\\_seam\\_rigidity\\_forcing.py](#)]). (iii) That clock-invariance is then *derived*, not assumed: the OS canonical transfer of  $\mu_4$ -symmetric RP data inherits the clock symmetry and its modular Hamiltonian commutes with it ([[verification/v446\\_seam\\_clock\\_invariance.py](#)]), so the residual drops from an operator condition to the structural  $\mu_4$ -symmetry of the seam RP data (QGEO.SYM.01). (iv) The number  $c_-=8$  is re-derived from a logically independent (topological) observable — the  $M=1$  collar has integer Chern  $C=+1$  (with  $C=0/C=-1$  controls) and the bulk–edge correspondence gives  $c_-=16 \cdot \frac{1}{2} \cdot 1 = 8$  ([[verification/v447\\_seam\\_edge\\_chern.py](#)]). (v) The one *external* MMST fact is pinned a further order: the edge state is quasi-free ( $G^2=G$ ) and the four-point cross-ratio Cauchy-converges to the conformal value  $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3}$  (against an off-scale gapped bulk control) ([[verification/v448\\_seam\\_edge\\_fourpoint.py](#)]). (vi) The MMST fact is pinned at the level of *uniformity*: two *independent* edge four-point cross-ratios converge to distinct conformal values ( $2+\sqrt{3}$  and 2) at

one  $N$ -independent  $1/N$  rate ( $|Q(N)-Q_\infty| \cdot N \leq 4$  for both) — the empirical hallmark the abstract theorem asserts ([[verification/v449\\_seam\\_mmst\\_uniform.py](#)]). (vii) The number  $c_- = 8$  is read a *third* independent way — the Calabrese–Cardy entanglement law gives  $c$  per Majorana  $= \frac{1}{2} (c_{\text{Dirac}} = 1.000$  vs a saturating gapped control), so  $c_- = 16 \cdot \frac{1}{2} = 8$  ([[verification/v450\\_seam\\_edge\\_entanglement.py](#)]). (viii) The edge CFT is named by its *operator content*: the Cardy finite-size spectrum reproduces the exact Ising primaries  $h_\sigma = \frac{1}{16}$  and  $h_\epsilon = \frac{1}{2}$  ( $L$ -independent), so the edge is the chiral Ising/Majorana minimal model  $M(3, 4)$  ([[verification/v451\\_seam\\_edge\\_cardy\\_tower.py](#)]). (ix) The  $(E_8)_1$  identity is pinned on the *torus*: the character  $\chi = E_4/\eta^8$  is  $S$ -invariant (one primary),  $T$ -multiplies by  $e^{-2\pi i/3} = e^{-2\pi i c/24}$  ( $c=8$ ) and is holomorphic ( $c \equiv 0 \pmod{8}$ ) ([[verification/v452\\_seam\\_e8\\_modular.py](#)]). (x) The last structural premise is *derived*: the four Gauss–Bonnet marks are  $\mu_4$  (roots of  $z^4-1$ ), the form basis is  $\mu_4$ -graded and the clock  $z \rightarrow iz$  fixes their cross-ratio, so the residual QGEO.SYM.01 of (iii) follows from marks  $= \mu_4$  plus the existing QGEO.REALIZE.01 — no new premise ([[verification/v453\\_seam\\_mu4\\_from\\_marks.py](#)]); the integer backbone of (iv)–(ix) ( $c_- = 8$ , the Chern integers, the order-3  $T$ -phase) is kernel-hardened in Lean axiom-free as FORM.SEAM.EDGE.CHERN.01. **None of these promotes the keystone**: the *only* thing that stays [O] is still the abstract continuum-existence theorem (v336); everything else is now corroborated, derived or kernel-checked.

**Target D — the CP-phase residual quantified [E] (not closed).** With the frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$ : pull vs  $\gamma_{\text{PDG}} = 65.7^\circ \pm 3.0^\circ$  is  $+0.98\sigma$  (survives). Audit, *not* promoted: the data central value sits  $0.07^\circ$  from the alternative coefficient reading  $\pi/3 + 2\lambda_C^2$  ( $|\mathbb{Z}_2|$  instead of  $N_{\text{fam}}$ ) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (REG.FREEZE.01). Decision threshold: the readings differ by  $\lambda_C^2 = 2.88^\circ$ , i.e.  $3\sigma$ -distinguishable once  $\sigma_\gamma \leq 0.96^\circ$ . The  $J$ -inversion is flagged magnitude-contaminated (source-scale  $s_{23}$  vs  $M_Z$ ). Ledger FLAV.CP.01.

**Target E — one flagged scale computed [C].** The reheating temperature is no longer a silent input:  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  [E] + standard physics give  $T_{\text{reh}} = 9.6 \times 10^9$  GeV and  $N_*(0.05/\text{Mpc}) = 51.4$  [C] (ledger COSMO.NSTAR.01); honestly recorded: this slow Higgs-channel point is  $A_s$ -disfavoured ( $-11.4\sigma$ ; the measured  $A_s$  requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band [50, 60] stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was merged (A), one was quantified (D), one input was computed (E).**

### Target D — the CP residual geometrically located (not closed) [E]/[C]

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. (i) **A channel typing** ([[verification/v227\\_degree\\_exponent\\_channel\\_split.py](#)]). The  $E_8$  split  $248 = 120 + 128$  is a magnitude/phase split: the exponents sum to  $120 = |R^+(E_8)|$  (the *magnitude* channel, where the (ratios, product)  $\Rightarrow v_{\text{geo}}$  bijection lives), the fundamental degrees sum to  $128 = \text{rank } E_8 \cdot \dim S^+$  (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong “ $S^-$  is dark matter” reading, now sharpened by v430: the other side  $S^-$  is matched  $\{2, 128\}$  structure, forced-disjoint from the unmapped  $\{12, 14, 18, 20, 24\}$ ). (ii) **A hexagonal modulus** ([[verification/v220\\_cp\\_hexagonal\\_modulus.py](#)]). The seam deck is the *square* modulus ( $j=1728, \mathbb{Z}/4$  clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus  $j=0$  ( $\tau=\rho=e^{i\pi/3}, \mathbb{Z}/6$ , CM by  $\mathbb{Z}[\omega]$ ). The frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$  has leading term  $\pi/3 = \arg \rho$ , the hexagonal CM phase; CP lives in the  $\mathbb{Z}/6$  phase *fiber* over the  $\mathbb{Z}/4$  seam, not in the deck. (iii) **An orientation** ([[verification/v225\\_dual\\_normal\\_frame.py](#)]). The dual normal frame  $d = a^\top R^{-1} = -\frac{1}{2}(1, 1, -2)$  (Nariai/horizon normal) and  $n = (5, -9, 6)$  (first-generation selector,  $n^\top R = (8, 0, 0)$ ) has oriented volume  $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$ , and rotating by the hexagonal unit gives  $\text{Im } \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3) \neq 0$  — CP sits precisely in the *orientation* of the normal frame, exactly where the magnitude bijection fails. All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry  $\delta$  is untouched). (iv) **One hexagonal unit, two sheets** ([[verification/v231\\_cp\\_mu6\\_phases.py](#)]). The reduction is sharper still: *both* CP phases are  $\mu_6$  powers of the single hexagonal CM unit  $\rho = e^{i\pi/3}$  —  $\delta_{\text{CKM}}^{\text{lead}} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the neutrino phase lattice) — with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  is the quark phase times the  $\mathbb{Z}_2$  sheet ( $\rho^3 = -1$ ); the frame orientations of (iii) are correspondingly sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). So Target D’s CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual  $3\lambda_C^2$  (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. (v) **The universal triality phase**

(`[verification/v233_cp_triality_phase.py]`). In the canonical  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation,  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight); since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of  $\mu_6$  power: the power choice of (iv) is removed, and  $\omega$  is the  $\mu_3$  factor of the order-30  $(2, 3, 5)$  Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C]. **(vi) The lock as a pre-registered kill test** (`[verification/v320_galois_cp_relation.py]`). Because  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  is now *forced* (one unit  $\rho = \zeta_6$ , the  $\mu_4$ -power  $\rho^4$ , the  $\mathbb{Z}_2$  sheet  $\rho^3 = -1$ ), it is a falsifiable cross-prediction, not a free input: a measured  $\delta_{\text{PMNS}}$  robustly away from  $240^\circ$  at DUNE/Hyper-K (beyond the sub-leading budget,  $> 3\sigma$ ) kills the Galois-CP organisation. The relation is [E], the physical phase reading stays [C], the kill test is [X].

#### Global look-elsewhere closure [E] (`[verification/v100_numerology_null_mc.py]`)

The per-target look-elsewhere traps recorded above (the  $\pi/3 + 2\lambda_C^2$  alternative in Target D, the  $\ln(46080)$  scale reading destroyed in `[verification/v99_koide_flow_time.py]`) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability  $\prod_i p_i = 10^{-25.8}$  over the 13 scored observables; all 94 500  $F_{U(1)}$  variants solved, exactly one hits CODATA ( $p_\alpha \leq 1.1 \cdot 10^{-5}$ ); total  $\leq 10^{-30.7}$ . The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT’s own hit count  $13 \rightarrow 9/6$ ; data shuffling  $\rightarrow 1.2$ ) prove it has power, and the census is stable under budget/window variation ( $< 2$  orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (tfpt\_3) and the module docstring.

#### Structural firewall + out-of-sample seed cross-validation [E] (`[verification/v305_witness_independence.py]`, `[verification/v306_seed_crossval.py]`)

The global null test answers “is the *scorecard* fishing?”; two companions answer the deeper adversarial worries “is the same integer counted many times as if independent?” and “is one shared seed dressed up as many predictions?”. **Witness economy** (`[verification/v305_witness_independence.py]`): every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the *single* anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive, so the recurring  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio, not many — 13 atoms from 2 free primitives ( $6.5\times$  compression made explicit). The  $E_8$  spine is genuinely overdetermined ( $\geq 3$  structurally distinct derivations sharing only leaf inputs), while the pure-seed readouts are single-witness, so their joint surprise is scored jointly by v100, never multiplied; the negative control shows  $\alpha^{-1}$ ’s “137” is *foreign* to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, v3). **Seed cross-validation** (`[verification/v306_seed_crossval.py]`): treating the axiom seed  $\varphi_0$  adversarially as a free dial and back-solving it from each of the five seed-grammar observables gives an error-weighted mean reproducing  $\varphi_0 = 1/(6\pi) + 48c_3^4$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  *out of sample* (the most-tensioned hold-out is  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ), and a data $\leftrightarrow$ formula shuffle explodes the cross-validation  $\chi^2$  by  $> 100\times$  (the test has power). Both stay [E] as structural/statistical checks, conditional on the same declared grammar. **No-golden-dynamics rule** (`[verification/v305_witness_independence.py]`): the golden ratio  $\phi = 2 \cos(\pi/5)$  lives in the *static* carrier field  $\mathbb{Q}(\sqrt{5})$  (the  $\mathbb{Z}/5$  hand; minimal polynomial  $x^2 - x - 1$ , discriminant 5 non-square), while every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6 = \text{sheet} \times \text{family}$  hand, e.g. the recovery factor  $(2/3)^6$ ) — so a dynamic rate “explained” by  $\phi$  is a cross-field false friend, and CP phases and recovery rates must run on the order-6 hand (the field firewall of v314/v315).

#### The reverse numerology audit, and what the golden ratio means [E]/[O] (`[verification/v354_e8_reverse_audit.py]`)

The witness-economy box answers “every readout maps onto  $E_8$  — is the *forward* map fishing?”. The deeper adversarial question runs the *other* way:  $E_8$  is a rich object, so how much of its structure maps onto *nothing*, and could the theory be quietly mining that richness? **The reverse audit.** Of  $E_8$ ’s

eight Casimir invariant degrees  $\{2, 8, 12, 14, 18, 20, 24, 30\}$  (exponents  $+1$ ), exactly *three* feed a *primary* physical-constant readout — degree 2 (the quadratic/metric), degree 8 (the rank  $\rightarrow c_3 = 1/(8\pi)$ ), degree 30 (the Coxeter number  $\rightarrow g_{\text{car}} = 5$  and the order-30 cycle) — and the other *five* carry no primary readout:  $\{12, 14, 18, 20, 24\}$ . (They do *coincide* with internal structural atoms —  $A_3$  roots,  $\dim G_2$ ,  $\det L$ ,  $|W(A_3)|$ , the anchor power sum — as **v66** records; but those per-degree coincidences are mostly *unforced*, the numerology temptation that the bandwidth search **v355** declines, see the next box.) So  $3/8$  are primary readouts,  $5/8$  hull overhead. **Economy, not promiscuity.** A *small fixed* set of  $E_8$  data (rank 8,  $h=30$ ,  $\det K=1$ , the  $D_5 \oplus A_3$  sub, the 16 half-spinor) generates the *entire* readout set (gauge group, three families, hypercharges,  $\alpha^{-1}$ , flavour, scales); the higher Casimirs are never reached into to fit. A numerological use of  $E_8$  would be *promiscuous* (mine the rich structure for matches); TFPT is *economical* (few invariants, many readouts) and never *reaches into* the higher Casimirs to fit; where those non-readout degrees do carry content it is *forced* internal structure (the spine and the flavour det-ladder, **v431**), not a mined coincidence — the anti-numerology signature is precisely this economy with a forced, *non-readout* remainder. **Why the golden ratio is not numerology.**  $\phi = 2 \cos(\pi/5)$  sits in exactly that unmapped part: it appears only in *internal* structure (the affine- $E_8$  attractor eigenvalue **v312**, the icosian lattice **v348**) and in no *frozen physical-constant* readout — the sole, *conditional* exception being the axion misalignment spine angle, where  $\theta_i = 3\pi/5$  is the regular-pentagon interior angle and  $\cos \theta_i = -1/(2\phi)$  ([**verification/v429\_axion\_pentagon\_phi.py**]; [**C**], branch-level, not a frozen prediction). Numerology means a number *fitted to an observation*;  $\phi$  is fitted to nothing — it is the algebraic signature of the 5-fold ( $h=30=2 \cdot 3 \cdot 5$ ), the carrier *announcing* that it is icosahedral, and even its one physical-ish appearance is forced geometry ( $\theta_i$  = the  $g_{\text{car}}$ -gon angle), not a match to a measured constant. **Honest residual.** The  $5/8$  hull overhead is exactly that:  $E_8$  is the minimal even-unimodular *hull* (consistency container,  $\det K=1$ ), not the world group, so the higher Casimirs are by-products of unimodular closure — where a future higher-invariant readout could live, or genuine over-capacity (now sharpened: the five non-readout degrees are *not* diffuse slack but the forced two-family ladder  $6 \cdot \text{spine} \sqcup \text{det-ladder}$ , **v431**). Stated plainly, not hidden.

**Bandwidth search of the unmapped region: the disciplined decline** [**E**]/[**C**]/[**O**]  
 ([**verification/v355\_e8\_unmapped\_bandwidth.py**])

The obvious next move — “then *search* those five degrees for hits we overlooked” — is itself the trap: a structure as rich as  $E_8$  returns a numerical match for almost anything. So the scan is run with a *strict discriminator*: a candidate counts only if it is a *forced* identity (a theorem about  $E_8$ ), never a coincidence. The honest yield is mostly a *decline*, and that decline is the result. **Forced, set-level (the genuinely overlooked structure — collective, not per-number).**  $\sum \deg = 128 = \dim S^+$ , the spinor half of  $248 = 120 + 128$  (a theorem:  $\sum \deg = \# \text{positive roots} + \text{rank} = 120 + 8$ ) — the invariant “budget” of  $E_8$  *equals the fermion content the carrier reads*;  $\sum \exp = 120 = \# \text{positive roots}$ ;  $\prod \deg = |W(E_8)| = 696\,729\,600$ ; and the exponents  $\{1, 7, 11, 13, 17, 19, 23, 29\}$  are exactly the  $\varphi(30) = 8$  totatives of 30 (**v223**), so the unmapped degrees’ exponents  $\{11, 13, 17, 19, 23\}$  are part of that forced set, not free. **Suggestive but not a theorem (the discriminator at work).**  $12 = h(E_6)$ ,  $18 = h(E_7)$ ,  $30 = h(E_8)$  are all degrees of  $E_8$  (the  $2T/2O/2I$  McKay ladder, **v219**); but “ $\deg(E_8)$  contains its subalgebras’ Coxeter numbers” is not a general theorem, so this appealing pattern is flagged [**C**], not claimed. **Declined — the numerology temptation.** The per-degree atom matches  $14 = \dim G_2$ ,  $20 = \det L$ ,  $24 = |W(A_3)| = 4!$ ,  $12 = |R(A_3)|$ ,  $18 = p_4(a)$  (which **v66** already flagged as fittable for 12, 18) each admit  $\geq 2$  readings from the frozen atoms with no forced selector, and the lone extra prime in  $|W(E_8)| = 2^{14}3^55^27$  “=” the scalaron exponent is one of many readings of 7 — all declined. **Reconciliation.** **v66** (the eight degrees coincide with the load-bearing integers, expected since  $E_8$  is the hull) and the reverse audit (**v354**: only 2, 8, 30 are primary readouts) are *both* right once forced set/subchain structure is separated from unforced per-degree coincidence. The search finds *no new forced physical hit*; the disciplined decline *is* the anti-numerology outcome.

**The “other side” of the seam vs the unmapped region** [**E**]/[**O**]  
 ([**verification/v430\_other\_side\_reverse\_audit.py**])

A sharp follow-up runs the audit through the *double cover*: the seam has an *other side* — the one-sided  $\mathbb{Z}_2$  collar and the conjugate half-spinor  $S^-$  — so *is the leftover where the second sheet lives?* The answer is a disciplined **negative**, the sheet/deck complement of **v354/v355**. **The deck is the degree-2 invariant (matched).** The one-sided cover contributes  $|\mathbb{Z}_2| = 2$  (the  $\frac{1}{2}$  in  $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$ , **v58/v216**), and  $2 = \min \deg(E_8)$  is the quadratic/metric — one of the *three matched* primary readouts  $\{2, 8, 30\}$ , never

one of the unmapped  $\{12, 14, 18, 20, 24\}$ . **The two sheets are the 128-spinor (matched, collective).**  $\dim S^+ = \dim S^- = 16$  and the spinor block is  $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$ , the spinor half of  $248 = 120 + 128$ ; the other side  $S^-$  enters 248 as the conjugate  $(\overline{16}, \overline{4})$  ( $128 = 2 \cdot 64$ ), *not* a spare singlet. **Forced-disjoint.** The sheet/deck integer set  $\{|\mathbb{Z}_2|, \dim S^+, \dim(S^+ \oplus S^-), 128\} = \{2, 16, 32, 128\}$  has *empty* intersection with the unmapped degrees  $\{12, 14, 18, 20, 24\}$ ; its only contact with the degree alphabet is the matched 2 (and the collective budget  $128 = \sum \deg$ ). **Disciplined decline [O].** Each unmapped degree, “explained” from the sheet atoms, needs an *unforced* coefficient and admits  $\geq 2$  readings — exactly the mining the v355 discriminator declines, so there is *no* forced sheet  $\rightarrow$  unmapped-degree identity. **Consequence.** This consolidates the  $S^-$ -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier  $E_8$  scan: the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five.

**The unmapped five are forced structure, not diffuse overhead: the degree ladder [E]/[C]**  
( [verification/v431\_e8\_degree\_ladder.py] )

The reverse audit (v354/v355) and its sheet/deck complement (v430) leave the five degrees  $\{12, 14, 18, 20, 24\}$  with no *primary* readout. The structural follow-up asks the sharper question — *are they diffuse hull overhead, or forced internal structure?* — and the answer is the latter, an *exact* two-family decomposition  $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \text{det-ladder}\{8, 14, 20\})$ . **The split is forced by 30.** The  $E_8$  exponents are exactly the  $\varphi(30) = 8$  totatives of  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  (v223), hence coprime to 6, hence  $\equiv \pm 1 \pmod 6$  — so the degrees ( $= \exp + 1$ ) occupy *only* the residue classes  $\{0, 2\} \pmod 6$ : two arithmetic progressions, forced by the prime content of  $30 = 2N_{\text{fam}}g_{\text{car}}$ . **The  $6k$  family is  $6 \times$  the spine.** Degrees  $\equiv 0 \pmod 6 = \{12, 18, 24, 30\}$ , divided by 6, give  $\{2, 3, 4, 5\}$  — the v91 spine  $\{e_3, p_0, e_1, e_2\}$  of the anchor  $a = (1, 1, 2)$ ;  $30 = 6g_{\text{car}}$  is the matched Coxeter readout, 12, 18, 24 were “unmapped.” **The  $6k+2$  family is the quadratic Casimir plus the flavour det-ladder.** Degrees  $\equiv 2 \pmod 6 = \{2, 8, 14, 20\}$ ; here  $\{8, 14, 20\} = (\det R, \det C, \det L)$  is the winding line  $\det M(s, 0) = 6s + 8$  of the flavour diamond (v135), with  $C = R + Q \text{diag}(1, 0, 0)$ , and 2 is the matched metric Casimir ( $8 = \det R$  is over-determined: also  $\text{rank } E_8$ ). **18 is not a holdout.**  $18 = 6 \cdot 3 = 6N_{\text{fam}}$  is the spine-family member  $6p_0$ , not on the det-ladder ( $6s + 8 = 18$  has no integer  $s$ ): it changes *family*, not status, so *no* genuinely orphan degree remains. **Special to  $E_8$ .** The clean  $\{0, 2\}$ -mod-6 split needs both  $6 \mid h$  and  $\text{exponents} = \text{totatives}(h)$ ; among  $\{A_4, D_5, E_6, E_7, E_8, F_4, G_2\}$  it holds for  $E_8$  but *fails* for  $E_6, E_7, D_5$ , so the decomposition is not a generic mod-6 triviality. **The honest v355 line.** The *arithmetic* decomposition is exact [E]; but the *functorial* claim “the  $6k+2$  Casimir stratum *is* the flavour sheet operators” needs a representation-theoretic map and stays [C] (degrees 12, 24 still admit two canonical readings each:  $6 \cdot \text{spine}$  vs  $|R(A_3)|/|W(A_3)|$ ). So this is a *structure* theorem — it reclassifies the  $5/8$  overhead of v354 to *zero* unfamilied degrees — not a forced functorial flavour map, and it closes no gate.

**The two structural integers and all of  $E_8$  are fixed by the degree multiset [E]**  
( [verification/v437\_e8\_degree\_joint.py] )

A final consolidation of the degree audit (of v6/v66/v355/v431), staying strictly inside the v354/v355 discipline — it adds *no* new per-degree coincidence, only one joint lock. **All of  $E_8$  is fixed by its degrees.** From  $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$  alone:  $\text{rank} = \# \deg = 8$ ,  $h = \max \deg = 30$ ,  $\# \text{positive roots} = \sum \exp = 120$ ,  $\# \text{roots} = 240$ ,  $\dim E_8 = 240 + 8 = 248$ ,  $|W(E_8)| = \prod \deg = 696\,729\,600$ ,  $\sum \deg = 128 = \dim S^+$  — every  $E_8$  integer the compiler uses is a function of the degrees. **The joint quadratic.** The two structural integers  $(g_{\text{car}}, N_{\text{fam}})$  are the *unique* root pair of  $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ : the sum of roots is the rank-fill  $\text{rank } E_8 = 8$  (v6) and the product is  $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$  ( $2g_{\text{car}}N_{\text{fam}} = h$ , v431), and  $\{3, 5\}$  is the unique positive-integer factor pair of 15 summing to 8 — so  $g_{\text{car}} = 5$  and  $N_{\text{fam}} = 3$  are forced *together* by two  $E_8$  degree-invariants, not one at a time. The three primary-readout degrees  $\{2, 8, 30\}$  are exactly  $\{\min \deg, \text{rank}, \max \deg\}$ , the invariants every simple Lie algebra distinguishes. Mirrored in Wolfram; closes no gate.

**External-review residual map [E]/[C] ( [verification/v182\_reviewer\_residual\_map.py] )**

A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: **(A)** the mapping  $2D \rightarrow 4D$  (“why does the dynamics read the masses as the Plücker norm 11?”), **(B)** the abstractness of UV

Table 1: The eight  $E_8$  Casimir invariant degrees, audited ([`verification/v66_e8_casimir_degrees.py`], [`verification/v354_e8_reverse_audit.py`], [`verification/v355_e8_unmapped_bandwidth.py`]; the sheet/deck complement [`verification/v430_other_side_reverse_audit.py`]; the forced two-family structure [`verification/v431_e8_degree_ladder.py`]). Only three feed a *primary* physical-constant readout; the other five carry no primary readout but are *not* diffuse overhead — they are the forced two families  $6 \cdot \text{spine}\{2, 3, 4, 5\}$  and the det-ladder  $\{8, 14, 20\}$ , the residue classes  $\{0, 2\} \bmod 6$  forced by  $h = 30$  (v431). The genuinely forced collective content:  $\sum \deg = 128 = \dim S^+$ ,  $\prod \deg = |W(E_8)|$ , exponents = totatives of 30.

| deg                                                                                                                                                   | exp | TFPT reading                 | forced?    | primary readout?      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------|------------|-----------------------|
| 2                                                                                                                                                     | 1   | $ \mathbb{Z}_2 $             | forced     | yes: metric           |
| 8                                                                                                                                                     | 7   | $\text{rank } E_8$           | forced     | yes: $c_3$            |
| 12                                                                                                                                                    | 11  | $h(E_6) =  R(A_3) $          | suggestive | no; 6·2 spine         |
| 14                                                                                                                                                    | 13  | $\dim G_2$                   | declined   | no; det $C$ ladder    |
| 18                                                                                                                                                    | 17  | $h(E_7) = p_4(a)$            | suggestive | no; 6·3 spine         |
| 20                                                                                                                                                    | 19  | $\det L$                     | declined   | no; det $L$ ladder    |
| 24                                                                                                                                                    | 23  | $ W(A_3)  = 4!$              | declined   | no; 6·4 spine         |
| 30                                                                                                                                                    | 29  | $h(E_8) = 2 \cdot 3 \cdot 5$ | forced     | yes: $g_{\text{car}}$ |
| <i>collective (forced):</i> $\sum \deg = 128 = \dim S^+$ , $\prod \deg =  W(E_8) $ , exp = totatives of 30                                            |     |                              |            |                       |
| <i>two families (forced, v431):</i> $\{0, 2\} \bmod 6$ ; $6k/6 = \text{spine } \{2, 3, 4, 5\}$ , $6k+2 = \{2\} \cup \text{det-ladder } \{8, 14, 20\}$ |     |                              |            |                       |

gravity, **(C)** the meta-look-elsewhere of the grammar choice, and **(D)** the Koide Möbius flow (non-integer flow time  $t \sim 2.84$ , missing QFT generator). None adds an *independent* open gap. **A** [**E**]/[**C**]: the readout integer is exact ( $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}}$ , the QBL boundary count, v174) and the  $2D \rightarrow 4D$  step is the holographic reduction, so  $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . **B** [**E**]: “gravity is the geometric readout of the boundary” *is* QGEO.SYM.01. **D** [**E**]/[**C**]:  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  is the Möbius group and the cusp flow lives on the deck  $\mathbb{P} \setminus \mu_4$ , so the Möbius *nature* is forced by the geometry; the non-integer  $t \sim 2.84$  only locates the physical point and the missing generator *is*  $F_{\text{transfer}}$ , so  $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (v84), minimal and over-determined (the anchor  $a = (1, 1, 2)$  lands in  $\geq 6$  mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay [**O**]/[**C**].

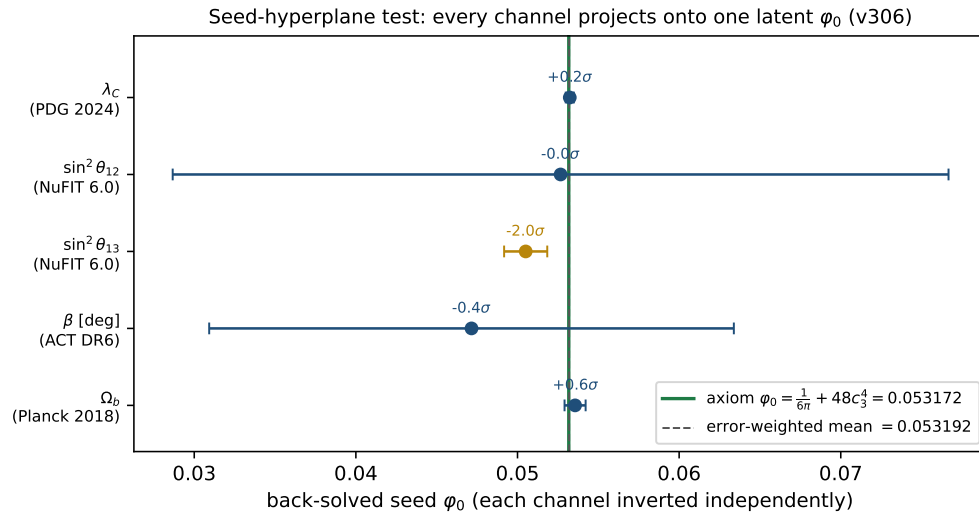


Figure 1: The seed-hyperplane test (`[verification/v306_seed_crossval.py]`): each scorecard observable is *independently* inverted to the latent seed  $\varphi_0$  (with its measurement error). All five project onto the one axiom value  $\varphi_0 = \frac{1}{6\pi} + 48c_3^4 = 0.053172$  (green), and their error-weighted mean reproduces it to 0.04% — one real seed, not five fits. The honest pressure point is  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  low, a *separate* carrier-trace channel, [v268](#)); a data $\leftrightarrow$ formula shuffle explodes the cross-validation  $\chi^2$  by  $> 100\times$ . [\[E\]](#).